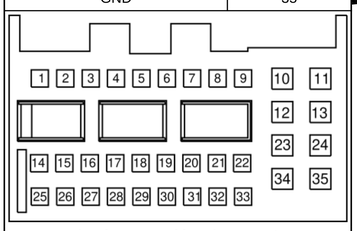
AVN_M05-B_F		
TE Connectivity: 2188701		
AMP_020060_35F	female	35-pin
MIC_RH_(+)		1
MIC_LH_(+)		2
NC		3
NC		4
NC		5
ILL_(+)		6
M-CAN_(HIGH)_1		7
NC		8
NC		9
PWR_12V_1		10
PWR_12V_2		11
GND		12
GND		13
MIC_RH_(-)		14
MIC_LH_(-)		15
NC		16
NC		17
NC		18
ILL_(-)		19
M-CAN_(LOW)_1		20
NC		21
PWR_12V_3		22
PWR_4		23
PWR_5		24
NC		25
NC		26
NC		27
NC		28
NC		29
NC		30
NC		31
PWR_12V_6		32
PWR_12V_7		33
GND		34
GND		35



view from rear of female connector

Additional components

11 x #14 (Crimp)

4 x #15 (Crimp)

4 x #16 (Crimp)

W3_b			
copper, TXL insulated	13x	0.3 mm²	0.15 m
AVN_M05-B_F:1:MIC_RH_(+)	1:BGRD	AVN_M05-B_MALE_MATE:1:MIC_RH_(+)	
AVN_M05-B_F:2:MIC_LH_(+)	2:BNRD	AVN_M05-B_MALE_MATE:2:MIC_LH_(+)	
AVN_M05-B_F:6:ILL_(+)	3:WHRD	AVN_M05-B_MALE_MATE:6:ILL_(+)	
AVN_M05-B_F:14:MIC_RH_(-)	4:BGBK	AVN_M05-B_MALE_MATE:14:MIC_RH_(-)	
AVN_M05-B_F:15:MIC_LH_(-)	5:BNBK	AVN_M05-B_MALE_MATE:15:MIC_LH_(-)	
AVN_M05-B_F:19:ILL_(-)	6:WHBK	AVN_M05-B_MALE_MATE:19:ILL_(-)	
AVN_M05-B_F:22:PWR_12V_3	7:OG	AVN_M05-B_MALE_MATE:22:PWR_12V_3	
AVN_M05-B_F:23:PWR_4	8:GNOG	AVN_M05-B_MALE_MATE:23:PWR_4	
AVN_M05-B_F:24:PWR_5	9:YEOG	AVN_M05-B_MALE_MATE:24:PWR_5	
AVN_M05-B_F:32:PWR_12V_6	10:PK	AVN_M05-B_MALE_MATE:32:PWR_12V_6	
S33:1:PWR_12V_7	11:RD	AVN_M05-B_MALE_MATE:33:PWR_12V_7	
AVN_M05-B_F:34:GND	12:BKBG	AVN_M05-B_MALE_MATE:34:GND	
AVN_M05-B_F:35:GND	13:BK	AVN_M05-B_MALE_MATE:35:GND	
Additional components			
0.15 m x #19 (Sleeve)			

AVN_M05-B_MALE_MATE	
wire-to-wire equiv. of TE 2188692	male 35-pin
1	MIC_RH_(+)
2	MIC_LH_(+)
3	NC
4	NC
5	NC
6	ILL_(+)
7	M-CAN_(HIGH)_2
8	NC
9	NC
10	PWR_12V_1
11	PWR_12V_2
12	GND
13	GND
14	MIC_RH_(-)
15	MIC_LH_(-)
16	NC
17	NC
18	NC
19	ILL_(-)
20	M-CAN_(LOW)_2
21	NC
22	PWR_12V_3
23	PWR_4
24	PWR_5
25	NC
26	NC
27	NC
28	NC
29	NC
30	NC
31	NC
32	PWR_12V_6
33	PWR_12V_7
34	GND
35	GND
Additional components	
11 x #14 (Crimp)	
4 x #15 (Crimp)	
4 x #16 (Crimp)	

W1_b			
copper, TXL insulated	1x	0.3 mm²	0.05 m
AVN_M05-B_F:33:PWR_12V_7	1:GYPK	S33:1:PWR_12V_7	

S33		
Splice	1-pin	
1	PWR_12V_7	1

W3_a			
copper, TXL insulated	4x	1.25 mm²	0.15 m
AVN_M05-B_F:10:PWR_12V_1	1:VT	AVN_M05-B_MALE_MATE:10:PWR_12V_1	
S11:1:PWR_12V_2	2:BU	AVN_M05-B_MALE_MATE:11:PWR_12V_2	
AVN_M05-B_F:12:GND	3:BKBN	AVN_M05-B_MALE_MATE:12:GND	
S13:1:GND	4:BK	AVN_M05-B_MALE_MATE:13:GND	

W1_a			
copper, TXL insulated	2x	1.25 mm²	0.05 m
AVN_M05-B_F:11:PWR_12V_2	1:BU	S11:1:PWR_12V_2	
AVN_M05-B_F:13:GND	2:BK	S13:1:GND	

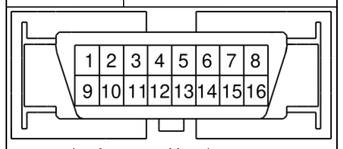
S11		
Splice	1-pin	
1	PWR_12V_2	1

S13		
Splice	1-pin	
1	GND	1

W3_c_twisted				
copper, TXL insulated, twisted pair	2x	0.3 mm²	+ S	0.15 m
S7:1:M-CAN_(HIGH)_2	1:GYOG	AVN_M05-B_MALE_MATE:7:M-CAN_(HIGH)_2		
S20:1:M-CAN_(LOW)_2	2:GYLB	AVN_M05-B_MALE_MATE:20:M-CAN_(LOW)_2		
Shield				

W2_a			
copper, TXL insulated	3x	0.3 mm²	0.3 m
S11:1:PWR_12V_2	1:LB	OBD2:9:PWR_12V_1	
S33:1:PWR_12V_7	2:GYPK	OBD2:1:PWR_12V_7	
S13:1:GND	3:BK	OBD2:5:GND	
Additional components			
0.3 m x #20 (Sleeve)			

OBD2	
Molex: 51115-1611	
OBD2	female 16-pin
1	PWR_12V_7
2	NC
3	M-CAN_(HIGH)_1
4	NC
5	GND
6	NC
7	NC
8	NC
9	PWR_12V_1
10	NC
11	M-CAN_(LOW)_1
12	M-CAN_(LOW)_2
13	M-CAN_(HIGH)_2
14	NC
15	NC
16	NC



view from rear of female connector

Additional components

7 x #17 (Crimp)

7 x #18 (Retainer)

W2_b_twisted				
copper, TXL insulated, twisted pair	2x	0.3 mm²	+ S	0.3 m
AVN_M05-B_F:7:M-CAN_(HIGH)_1	1:GYRD	SMH1:1:M-CAN_(HIGH)_1		
AVN_M05-B_F:20:M-CAN_(LOW)_1	2:GYBU	SML1:1:M-CAN_(LOW)_1		
Shield				

SMH1		
Splice	1-pin	
1	M-CAN_(HIGH)_1	1

SML1		
Splice	1-pin	
1	M-CAN_(LOW)_1	1

W2_d_twisted				
copper, TXL insulated, twisted pair	2x	0.3 mm²	+ S	0.05 m
SMH1:1:M-CAN_(HIGH)_1	1:GYRD	OBD2:3:M-CAN_(HIGH)_1		
SML1:1:M-CAN_(LOW)_1	2:GYBU	OBD2:11:M-CAN_(LOW)_1		
Shield				

W2_c_twisted				
copper, TXL insulated, twisted pair	2x	0.3 mm²	+ S	0.3 m
S7:1:M-CAN_(HIGH)_2	1:GYOG	SMH2:1:M-CAN_(HIGH)_2		
S20:1:M-CAN_(LOW)_2	2:GYLB	SML2:1:M-CAN_(LOW)_2		
Shield				

SMH2		
Splice	1-pin	
1	M-CAN_(HIGH)_2	1

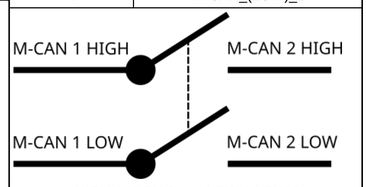
SML2		
Splice	1-pin	
1	M-CAN_(LOW)_2	1

W2_e_twisted				
copper, TXL insulated, twisted pair	2x	0.3 mm²	+ S	0.05 m
SMH2:1:M-CAN_(HIGH)_2	1:GYOG	OBD2:13:M-CAN_(HIGH)_2		
SML2:1:M-CAN_(LOW)_2	2:GYLB	OBD2:12:M-CAN_(LOW)_2		
Shield				

W2_f_twisted				
copper, TXL insulated, twisted pair	2x	0.3 mm²	+ S	0.05 m
SMH1:1:M-CAN_(HIGH)_1	1:GYRD	switch:1:M-CAN_(HIGH)_1		
SML1:1:M-CAN_(LOW)_1	2:GYBU	switch:2:M-CAN_(LOW)_1		
Shield				

W2_g_twisted				
copper, TXL insulated, twisted pair	2x	0.3 mm²	+ S	0.05 m
SMH2:1:M-CAN_(HIGH)_2	1:GYOG	switch:3:M-CAN_(HIGH)_2		
SML2:1:M-CAN_(LOW)_2	2:GYLB	switch:4:M-CAN_(LOW)_2		
Shield				

switch		
Alcoswitch: TSS21KGPC	TE Connectivity: 1-1437579-6	
double pole single throw	latching 5VDC	4-pin
1	M-CAN_(HIGH)_1	
2	M-CAN_(LOW)_1	
3	M-CAN_(HIGH)_2	
4	M-CAN_(LOW)_2	



HIGH 1-HIGH 2 and LOW 1-LOW 2

Additional components

2 x #1 (5.08mm pitch wire-to-board receptacle)

4 x #13 (Crimp)

1. all wire connections crimped or screw connectors, not soldered
2. all connectors to oem spec as in drawing C-2188701 and SD-51115-001
3. all crimping for 2188701 and male mate connectors according to spec 411-61029 except as necessary to mate 2 wires
4. all crimping for 51115-1611 according to spec in drawing SD-51115-001
5. wrap all wires in PET or nylon braided sleeves to 30mm from connectors
5. a) wrap all W2 wires in one sleeve
   b) wrap all W3 wires and switch in one sleeve
   c) cut hole for access to switch
7. use of combined 73-pin male mate (wire-to-wire equivalent of 2188692) is acceptable if it the most
   economical option; use of 35-pin-only connector is preferred if cost is the same or better
8. wire lengths are +/- 30%
9. any splice type suitable for long-term automotive applications is acceptable
10. substitution of parts for cost is acceptable as long as replacements meet original specification
11. simplifications for cost/ease of manufacturing may be discussed
12. wire colors are for clarity and not required
13. all CAN lines shielded, twisted pairs

Name:	Date:
Hyundai/Kia M-CAN to OBD dongle	2026-03-13
Drawing Number:	**Drawn by:**
HKMOBD_SR_ES-002	Tyler Harvey